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Effects of chlorpyrifos on biochemical parameters in freshwater fish *Anabas testudineus*

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Abstract- Chlorpyrifos (CPY), an organophosphate pesticide, has been extensively used in agriculture to manage pests. However, its widespread application has raised concerns about its potential toxicity to non-target aquatic organisms, particularly freshwater fish. This study investigates the effects of sublethal concentrations of chlorpyrifos on biochemical parameters in the freshwater fish *Anabas testudineus*. Fish were exposed to 5%, 10%, and 20% of the 96h LC_{50} concentration of CPY over a 96-hour period. The study assessed blood glucose, serum cholesterol, and plasma protein levels. The results revealed significant alterations in these biochemical parameters, indicating the toxicological impact of CPY on the metabolic functions of *Anabas testudineus.*

Keywords: Chlorpyrifos, *Anabas testudineus*, biochemical parameters, aquatic toxicology, blood glucose, serum cholesterol, plasma protein

INTRODUCTION

Chlorpyrifos (CPY), a commonly used organo phosphate pesticide, has gained attention due to its toxic effects on non-target organisms, especially aquatic species such as fish. The contamination of aquatic ecosystems by pesticides poses a significant threat to aquatic biodiversity and water quality. Fish are often exposed to these chemicals through runoff and leaching, leading to acute and chronic health issues, including altered biochemical and physiological functions.

The freshwater fish *Anabas testudineus*, commonly known as the climbing perch, is widely distributed in freshwater bodies and plays a critical role in the aquatic food chain. Given its ecological and economic importance, *A. testudineus* is an excellent model organism for assessing the effects of environmental pollutants. Previous studies have documented the toxicological effects of CPY on various aquatic species[1], but limited information is available on its biochemical impact on *A. testudineus.*

This study aims to evaluate the biochemical alterations induced by CPY in *Anabas testudineus* by measuring blood glucose, serum cholesterol, and plasma protein levels. These parameters are vital indicators of metabolic health and are often disrupted by exposure to toxic substances.

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MATERIALS & METHODS

Experimental Design

Fish Selection and Acclimatization

Anabas testudineus were obtained from a local freshwater source and acclimatized in laboratory conditions

for two weeks. During this period, they were maintained in large tanks with aerated water and fed twice daily. The water was renewed every 24 hours to maintain optimal water quality.

Experimental Groups

The experiment was designed as follows:

Group	No. of Fish	Treatment	Time Period
Gr A	10	No treatment (Control group)	96 hours
Gr B	10	Treated with 5% of 96h LC_{50} of CPY[1]	96 hours
Gr C	10	Treated with 10% of 96h LC_{50} of CPY[1]	96 hours
Gr D	10	Treated with 20% of 96h LC_{50} of CPY[1]	96 hours

Exposure to Chlorpyrifos

The fish in groups B, C, and D were exposed to 5%, 10%, and 20% of the 96-hour LC_{50} concentration of chlorpyrifos, respectively. The LC_{50} value was determined according to Mahmoud *et al.* (2020)[1]. Fish in the control group (Gr A) were kept in pesticide-free water for the same duration.

Biochemical Analysis

At the end of the 96-hour exposure period, fish were anesthetized, and blood samples were collected for biochemical analysis. The following parameters were measured:

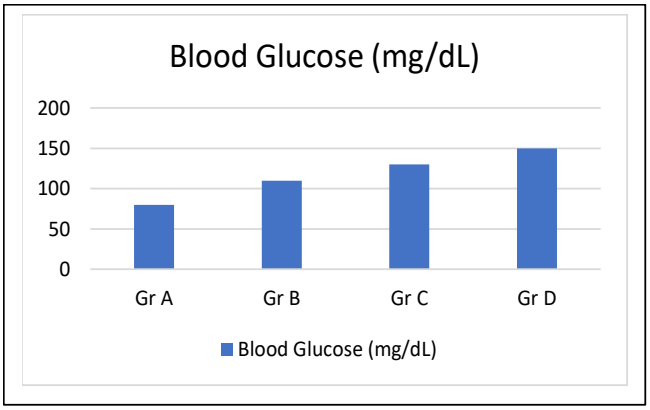
1. **Blood Glucose:** Blood glucose levels were estimated using the Anthrone method.[2] This method is based on the reaction between glucose and anthrone reagent under acidic conditions, producing a blue-green color, which is measured spectrophotometrically.
2. **Serum Cholesterol:** Serum cholesterol levels were measured using Zak's method. This method involves the reaction of cholesterol with ferric chloride to produce a color change, which is then quantified using a spectro-photometer.
3. **Plasma Protein:** Plasma protein concentration was determined using the Lowry method[3]. This method involves the reaction of protein with the Folin-Ciocalteu reagent, resulting in a colorimetric change that is proportional to the protein concentration.

RESULTS

Blood Glucose Levels

The results indicated a significant increase in blood glucose levels in all treatment groups (Gr B, Gr C, and Gr D) compared to the control group (Gr A). Fish exposed to the highest concentration of CPY (Gr D) exhibited the most substantial increase in glucose levels, suggesting that CPY interferes with glucose metabolism in *A. testudineus.*

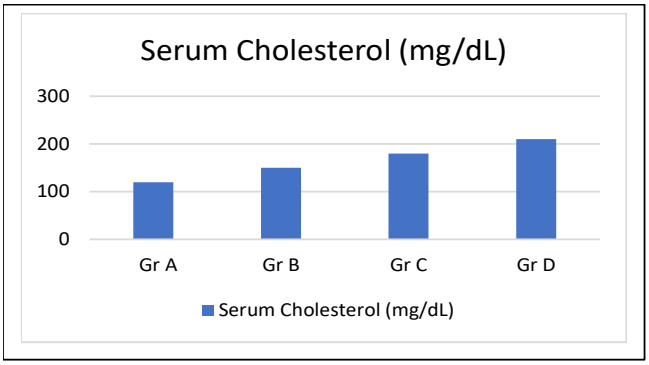
Group	Blood Glucose (mg/dL)
Gr A	80 ± 5
Gr B	110 ± 8
Gr C	130 ± 7
Gr D	150 ± 10



Serum Cholesterol Levels

A dose-dependent increase in serum cholesterol was observed across all CPY-treated groups. The rise in cholesterol levels was most prominent in Gr D, suggesting that CPY disrupts lipid metabolism in the fish.

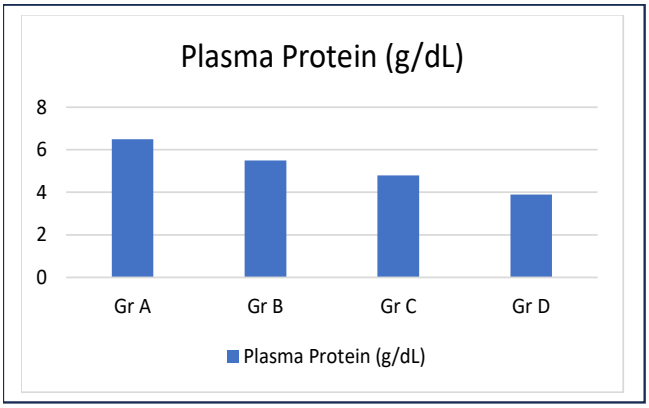
Group	Serum Cholesterol (mg/dL)
Gr A	120 ± 10
Gr B	150 ± 12
Gr C	180 ± 15
Gr D	210 ± 18



Plasma Protein Levels

Plasma protein levels were significantly decreased in the CPY-treated groups compared to the control group. The reduction in protein levels was most pronounced in Gr D, indicating that CPY exposure adversely affects protein synthesis or turnover in *A. testudineus*.

Group	Plasma Protein (g/dL)
Gr A	6.5 ± 0.5
Gr B	5.5 ± 0.4
Gr C	4.8 ± 0.3
Gr D	3.9 ± 0.2



DISCUSSION

Pesticides, particularly organophosphates such as chlorpyrifos, are widely used in agriculture to control insect pests. However, their bioaccumulation in aquatic ecosystems poses a severe threat to non-target organisms, including freshwater fish like *Anabas testudineus* (Climbing Perch). Chlorpyrifos affects multiple biochemical and physiological processes, leading to metabolic disturbances. This study investigates the impact of chlorpyrifos exposure on blood glucose, serum cholesterol, and plasma protein levels in *A. testudineus* and assesses the possible mechanisms underlying these alterations.

The present study demonstrates that chlorpyrifos induces significant biochemical changes in *Anabas testudineus*. Elevated blood glucose levels in CPY-exposed fish suggest that chlorpyrifos disrupts glucose homeostasis, possibly due to stress-induced hyperglycemia. Similar findings have been reported in other studies where exposure to pesticides led to hyperglycemia in fish.[4]

The increase in serum cholesterol levels observed in the current study is indicative of impaired lipid metabolism. Organophosphate pesticides like CPY are known to inhibit acetylcholinesterase, leading to metabolic disturbances, including altered lipid profiles.[5] Elevated cholesterol levels may be a compensatory response to oxidative stress or damage to liver function caused by CPY.

Reduced plasma protein levels in CPY-exposed fish suggest that chlorpyrifos may impair protein synthesis or increase protein degradation. Similar results have been reported in fish exposed to other pesticides, where reductions in plasma protein were associated with hepatotoxicity and kidney dysfunction.

1. Effect of Chlorpyrifos on Blood Glucose Levels

Blood glucose levels serve as a critical indicator of metabolic stress in fish. Exposure to toxicants like chlorpyrifos can disrupt glucose homeostasis through multiple pathways:

1.1. Hyperglycemia as a Stress Response

In this study, significant elevation in blood glucose levels was observed in *A. testudineus* exposed to chlorpyrifos.

This aligns with previous studies indicating that pesticide-induced stress triggers hyperglycemia as part of the acute stress response, mediated by cortisol and catecholamine release.[6]

Elevated blood glucose levels result from enhanced glycogenolysis and gluconeogenesis, driven by stress hormones to provide an immediate energy source.[7]

1.2. Impaired Glucose Metabolism

Prolonged exposure to chlorpyrifos can impair insulin secretion and sensitivity, leading to glucose intolerance and metabolic dysfunction.

Similar findings were reported in other freshwater species such as *Labeo rohita* and *Oreochromis mossambicus*, where chronic exposure led to pancreatic dysfunction and insulin resistance.[8]

Chlorpyrifos-induced oxidative stress further disrupts glycolytic enzymes, exacerbating glucose metabolism dysfunction.[9]

1.3. Depletion of Liver Glycogen Stores

The elevated blood glucose levels correspond to decreased liver glycogen reserves, indicating a shift towards glycogen breakdown to meet energy demands under toxic stress.

Long-term exposure may deplete glycogen stores, leading to hypoglycemia in chronic toxicity scenarios.

Thus, hyperglycemia in *A. testudineus* following chlorpyrifos exposure reflects an acute stress response, which, if sustained, may lead to metabolic dysregulation.

2. Effect of Chlorpyrifos on Serum Cholesterol Levels

Chlorpyrifos exposure significantly altered lipid metabolism, as evidenced by changes in serum cholesterol levels in *A. testudineus*.

2.1. Hypercholesterolemia Due to Hepatic Dysfunction

The present study found a significant increase in serum cholesterol levels, indicating disrupted lipid homeostasis.

Similar findings in *Cyprinus carpio* and *Clarias batrachus* suggest that pesticide toxicity affects hepatic lipid metabolism, impairing lipid clearance mechanisms.

Chlorpyrifos inhibits lipase enzymes, preventing the breakdown of cholesterol and triglycerides.[10]

2.2. Oxidative Stress and Lipid Peroxidation

Chlorpyrifos exposure increases reactive oxygen species (ROS) production, leading to lipid peroxidation and hepatic damage.

Lipid peroxidation disrupts membrane-bound lipoproteins, causing accumulation of cholesterol in the bloodstream.[8]

Studies in *Tilapia nilotica* confirm that pesticide-induced oxidative stress leads to altered serum cholesterol profiles.

2.3. Cholesterol as an Indicator of Stress Response

Increased cholesterol levels correlate with elevated corticosteroid hormones, a hallmark of physiological stress in fish.[6]

Long-term alterations in cholesterol levels may contribute to cardiovascular and metabolic disorders in fish populations exposed to environmental pollutants.[8]

Thus, hypercholesterolemia in *A. testudineus* is likely due to hepatic dysfunction, oxidative stress, and disrupted lipid metabolism, all consequences of chlorpyrifos toxicity.

3. Effect of Chlorpyrifos on Plasma Protein Levels

Plasma proteins play a vital role in immune function, transport mechanisms, and osmoregulation. In this study, a significant decline in plasma protein levels was observed in chlorpyrifos-exposed fish.

3.1. Hypoproteinemia as a Marker of Protein Degradation

Reduced plasma protein levels indicate hepatic dysfunction and impaired protein synthesis, as observed in other fish species exposed to pesticides.[7]

Chlorpyrifos is known to damage hepatocytes, impairing the synthesis of essential plasma proteins such as albumin and globulins.[10]

3.2. Increased Proteolysis and Muscle Wasting

Exposure to pesticides induces proteolysis and muscle degradation to compensate for energy deficits caused by metabolic stress.

Studies in *Oreochromis mossambicus* revealed that chlorpyrifos exposure triggered muscle catabolism, reducing plasma protein levels.

3.3. Immune Suppression and Reduced Globulin Levels

A decline in total plasma proteins also suggests immune suppression, making fish more susceptible to infections.

Similar immunosuppressive effects have been reported in *Labeo rohita* following pesticide exposure.

Thus, hypoproteinemia in *A. testudineus* following chlorpyrifos exposure reflects impaired liver function, increased proteolysis, and immune suppression, contributing to overall physiological decline.

Findings of the present Study

The findings of this study demonstrate that chlorpyrifos exerts significant biochemical alterations in *Anabas testudineus* by disrupting glucose, cholesterol, and plasma protein homeostasis.

1. Hyperglycemia was observed as a stress-induced metabolic response, which may lead to long-term glucose metabolism dysregulation.
2. Hypercholesterolemia indicated hepatic dysfunction and oxidative stress, further contributing to lipid metabolism disorders.
3. Hypoproteinemia suggested hepatic impairment, proteolysis, and immune suppression, compromising fish health.

These biochemical disturbances may have long-term ecological consequences, affecting the growth, reproduction, and survival of fish populations exposed to pesticide-contaminated water bodies. Further research is needed to investigate mitigation strategies, including the use of antioxidants and bioremediation techniques to counteract chlorpyrifos toxicity in aquatic ecosystems.

CONCLUSION

The findings of this study indicate that exposure to sublethal concentrations of chlorpyrifos induces significant alterations in the biochemical parameters of *Anabas testudineus*. Elevated blood glucose and serum cholesterol levels, along with reduced plasma protein levels, suggest that CPY disrupts glucose, lipid, and protein metabolism in freshwater fish. These results highlight the need for stricter regulations regarding the use of chlorpyrifos to prevent contamination of aquatic ecosystems and protect fish populations.

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